

# Nicolás Alejandro Comay

Psychologist, PhD in Neuroscience

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## Education

**PhD in Neuroscience**, National University of Córdoba (UNC-Argentina). 2021-2026

Supervisor: Dr. Pablo Barttfeld. Co-supervisor: Dr. Guillermo Solovey.

**Degree in Psychology**, National University of Córdoba (UNC-Argentina). 2016-2020

Recipient of the 'University Award' for academic excellence (highest GPA among on-time graduates).

## Publications

**Comay, N.A.**; Solovey G.; Barttfeld, P. Confidence phenotypes: a unified computational account of value and decision certainty in reinforcement learning. *eLife*, 15:RP111820. 2026

**Comay, N.A.**; Solovey, G.; Barttfeld, P. Iconic vs. working memory metacognition to evaluate the richness of perception: a registered report. *Royal Society Open Science*, 12(11), 231805. 2025

**Comay, N.A.**; Solovey, G.; Barttfeld, P. Decisions are based on less information than metacognitive judgments in multialternative contexts. *Journal of Experimental Psychology: Learning, Memory and Cognition*, 52(7), 1080–1093. 2025

Barttfeld, P.; **Comay, N.A.**; Embon, I.; Solovey, G. Pre- and post-decision signals of certainty in changing minds. *Trends in Neurosciences*, 48(12), 935–937. 2025

Zapata, J. M.<sup>+</sup>, **Comay, N.A.**<sup>+</sup>, Taricco, G., Barttfeld, P., Solovey, G., Saal, A., & Ahumada, J. V. Metacognitive sensitivity on the Iowa Gambling Task reveals awareness as a necessary condition for advantageous performance. *Experimental Psychology*, 71(6), 343–351. <sup>+</sup>joint first authors. 2025

**Comay, N.A.**, Della Bella, G., Lamberti, P., Sigman, M., Solovey, G., & Barttfeld, P. The presence of irrelevant alternatives paradoxically increases confidence in perceptual decisions. *Cognition*, 234, 105377. 2023

## Scholarships, awards and research funding

### FELLOWSHIPS

*PhD fellowship*. National Scientific and Technical Research Council of Argentina (CONICET). 2021

*Stimuli to Scientific Vocations scholarship*. Undergraduate scholarship from the National Interuniversity Council of Argentina. 2020

### AWARDS

*Dr. Pedro Pavesi award* for best presentation at the conference The Future of Decision-Making Sciences (Center of Rationality, Action and Decision – Economics School – University of Buenos Aires). 2022

*University Award* from the National University of Córdoba for academic excellence (highest GPA among on-time graduates). 2021

### RESEARCH FUNDING

Pre-stimulus neural activity and conscious perception: a theory-informed investigation using report and no-report fMRI paradigms. *Consciousness Commons 2026 prize*. USD 8,960. 2026

Joint project by four PhD candidates; co-recipients of the award.

*Incubator project prize*, First Annual Summer School on Consciousness and Metacognition (University College London and Université de recherche Paris Sciences et Lettres). GBP 250. 2021

Joint project by four PhD candidates; co-recipients of the award.

## TRAVEL GRANTS FOR ACADEMIC EVENTS

*Travel grant* to attend the 2025 Multi-disciplinary Conference on Reinforcement Learning and Decision Making (Trinity College Dublin, Ireland) **2025**

*Travel grant* to attend the Neuroscience of Consciousness Winter School (Canadian Institute for Advanced Research, CIFAR). **2024**

*Travel grant* to attend the RIKEN Center for Brain Science Summer School (Tokyo, Japan) **2023**

*EMBO travel grant* to attend the Barcelona Summer School for Advanced Modeling of Behavior (Centre de Recerca Matemàtica – Barcelona, Spain). **2022**

## Research experience

### AS A GRADUATE STUDENT

Three-month research internship at the Laboratory for Imagination and Executive Functions led by Dr. Kentaro Miyamoto (*RIKEN Center for Brain Science* – Hirosawa, Wako, Saitama 351-0106, Japan). **2024**

Two-month research internship at the Laboratory for Imagination and Executive Functions led by Dr. Kentaro Miyamoto as part of the *RIKEN Center for Brain Science Summer School Program* (Hirosawa, Wako, Saitama 351-0106, Japan). **2022**

Ph.D. candidate in Neuroscience at the Psychological Research Institute (National Scientific and Technical Research Council – National University of Córdoba, Argentina). **2021-present**

### AS AN UNDERGRADUATE STUDENT

Undergraduate research assistant, supported by the *Stimuli to Scientific Vocations Scholarship* from the National Interuniversity Council of Argentina. Project: *Effects of pre- and/or postnatal ethanol exposure on the modulation of the opioid system in neonatal rats*. **2020-2021**

Thesis in Psychology (Psychological Research Institute, UNC-CONICET). Title: *Confidence in perceptual decisions with multiple alternatives*. Qualification: 10/10. Supervisor: Dr. Pablo Barttfeld. Co-supervisor: Dr. Guillermo Solovey. **2020**

Undergraduate research assistant. Project: *Neuroprotection facilitated by Omega 3 in two cognitive impairment models*. Psychological Research Institute (UNC-CONICET, Argentina). **2019**

## Presentations

### ORAL PRESENTATIONS & TALKS

*Confidence phenotypes: a unified computational account of value and decision certainty in reinforcement learning* (**Comay, N.A.**; Solovey, G.; Barttfeld, P.). 48th Annual Meeting of the Cognitive Science Society. Rio de Janeiro, Brasil. **2026**

*Iconic vs. working memory metacognition to evaluate the richness of perception: a registered report* (**Comay, N.A.**; Solovey, G.; Barttfeld, P.). 30th annual meeting of the Association for the Scientific Study of Consciousness. Santiago, Chile. **2026**

*Bayesian approach to perceptual decision making and confidence* (**Comay, N.A.**; Solovey, G.; Barttfeld, P.). Bayesian Plurinational Congress. Santiago del Estero, Argentina. **2023**

*Computational modeling of behavior* (**Comay, N.A.**; Solovey, G.; Barttfeld, P.). V International Psychology Congress and VIII National Psychology Congress “Science and Profession” (UNC, Argentina). **2023**

*Deviations from Bayesian optimality in the evaluation of perceptual decisions* (**Comay, N.A.**; Della Bella, G.; Lamberti, P.; Sigman, M.; Solovey, G.; Barttfeld, P.). The future of decision-making sciences conference (Center of Rationality, Action and Decision - Economics School – University of Buenos Aires). **2022**

*Implausible alternatives paradoxically increase confidence in a perceptual decision* (**Comay, N.A.**; Della Bella, G.; Lamberti, P.; Sigman, M.; Solovey, G.; Barttfeld, P). XXXVI Annual Meeting of the Argentinian Society for Neuroscience Research. **2021**

## WORKSHOPS & SYMPOSIA

*Thinking how we think: actual controversies in metacognition and experimental designs.* (Embon, I.<sup>+</sup>; Lillo, M.<sup>+</sup>; **Comay, N.A.**<sup>+</sup>; Aguirre, J.<sup>+</sup>). II International Meeting Frontiers in Neurosciences (Universidad Nacional Autónoma de México, UNAM). <sup>+</sup>equal contribution. **2023**

*Reflecting upon our own thoughts: recent studies on metacognition in vigilance, personality and decision making* (**Comay, N.A.**<sup>+</sup>; Aguirre, J.; Embon, I.; Gerbaudo, M.A.; Lillo, M.). XIX National Meeting and VIII International Meeting of the Argentinean Association for Behavioral Sciences.<sup>+</sup>symposium coordinator **2023**

## POSTER PRESENTATIONS

*Bayesian-like computations explain human confidence in values and decisions during learning* (**Comay, N.A.**, Solovey, G.; Barttfeld, P.). 2025 Multi-disciplinary Conference on Reinforcement Learning and Decision Making. Trinity College Dublin, Ireland. **2025**

*Computational modeling of uncertainty in reinforcement learning* (**Comay, N.A.**, Solovey, G.; Barttfeld, P.). I Congreso Argentino de Ciencias Cognitivas. Universidad Torcuato Di Tella, Buenos Aires. **2024**

*Implausible alternatives paradoxically increase confidence in a perceptual decision.* (Comay, N.A.; Della Bella, G.; Lamberti, P.; Sigman, M.; Solovey, G.; Barttfeld, P.) Computational Cognition Workshop (Osnabrück, Germany). **2021**

*Effect of subliminal feedback on metacognition in a visual perception task.* (Zapata, J.M.<sup>+</sup>; **Comay, N.A.**<sup>+</sup>; Della Bella, G.<sup>+</sup>). XXXV Annual Meeting of the Argentinian Society for Neuroscience Research. <sup>+</sup>equal contribution. **2020**

*Deterioration of grid cells and loss of spatial orientation in Alzheimer's Disease* (Cabrera, V.<sup>+</sup>; **Comay, N.A.**<sup>+</sup>; Maza, P.<sup>+</sup>; Vietéz Salvá, E.<sup>+</sup>; Macchione, A.F.<sup>+</sup>). V International Psychology Congress and VII National Psychology Congress "Science and Profession". <sup>+</sup>equal contribution. **2020**

## Teaching experience

### GRADUATE-LEVEL

Course instructor and coordinator, *Mathematical and computational modelling of behaviour*, Master's Program in Research Methodology in Behavioural Sciences. **2026**  
National University of Córdoba

Teaching assistant, *A practical introduction to cognitive science.* **2023**  
National University of Córdoba

### UNDERGRADUATE-LEVEL

Teaching assistant, *Tools for data analysis in cognitive neuroscience* **2023, 2026**  
National University of Córdoba

Undergraduate teaching assistant, *Neurophysiology and Psychophysiology* **2020-2022**  
National University of Córdoba

Undergraduate teaching assistant, *Experimental Psychobiology* **2019-2021**  
National University of Córdoba

## Human resources training

### UNDERGRADUATE STUDENTS SUPERVISED IN THE PSYCHOLOGY PROGRAM

Elias Franco Parras (UNC, Argentina). **2025-2026**

|                                                                                 |           |
|---------------------------------------------------------------------------------|-----------|
| Juan Taravella & Lucia Molina (UNC, Argentina).                                 | 2024-2025 |
| Gaspar Taricco (UNC, Argentina). <i>Co-supervised</i> .                         | 2024-2025 |
| Bautista Squire (UNC, Argentina). <i>Co-supervised</i> .                        | 2024-2025 |
| Agostina Presuttari & Camila Abud (UNC, Argentina).                             | 2024-2025 |
| Vera Candusso & Julieta Irazuzta (UNC, Argentina).                              | 2024-2025 |
| Benjamin Sanchez (UNC, Argentina). <i>Co-supervised</i> .                       | 2023-2024 |
| Camila Inés Amado (UNC, Argentina).                                             | 2023-2024 |
| Candelaria Nicolas & Florencia Ledesma (UNC, Argentina). <i>Co-supervised</i> . | 2023-2024 |

## Peer review and academic service

### PEER REVIEW FOR JOURNALS

Nature communications; Nature Machine Intelligence; NeuroImage: Reports; Scientific Reports; npj Artificial Intelligence; Frontiers in Psychology

### UNDERGRADUATE THESES EVALUATED

Evaluation of five undergraduate theses in Psychology at the National University of Córdoba.

### INSTITUTIONAL SERVICE

Member of the ethics committee for the Use and Care of Laboratory Animals at the Psychological Research Institute (UNC-CONICET) **2022-present**

## Skills

Programming (R, MATLAB).

Computational modelling.

Advanced statistical analysis.

Data analysis and visualisation.

fMRI: FSL (FMRIB Software Library) & fMRIPrep.

EEG: EEGLAB.

## Selected summer schools & courses

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|-----------------------------------------------------------------------------------------------------------------------|------|
| Neuroscience of Consciousness Winter School (Canadian Institute for Advanced Research)                                | 2024 |
| RIKEN Center for Brain Science Summer School (RIKEN CBS, Japan)                                                       | 2023 |
| Barcelona Summer School for Advanced Modeling of Behavior (Centre de Recerca Matemàtica, Spain)                       | 2022 |
| Statistics applied to health sciences (UNC, Argentina)                                                                | 2022 |
| First Annual Summer School on Consciousness and Metacognition (University College London & Paris Sciences et Lettres) | 2021 |
| Computational Neuroscience (UNSAM, Argentina)                                                                         | 2021 |
| Python programming (UNSAM, Argentina)                                                                                 | 2021 |
| Decision making (UBA, Argentina)                                                                                      | 2020 |
| Behavioral economics (UBA, Argentina)                                                                                 | 2020 |

## References

**Pablo Barttfeld, PhD**

Independent Researcher at CONICET

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**Guillermo Solovey, PhD**

Independent Researcher at CONICET

Universidad Torcuato Di Tella (UTDT), Argentina

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